

Quantum gates and algorithms in qg units: Hamming-weight conservation, readout classes and the local radius

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Abstract

We write fifteen standard quantum gates and fourteen standard quantum algorithms in qg units [1]. An n -qubit state is described by its qg values, $qg_P = \langle P \rangle$ for every Pauli string P , and a gate U acts as $qg'_P = qg_{U^\dagger P U}$. With this one rule each gate becomes a table of qg transformations and each algorithm a readout identity: which qg values hold the answer. The formulation is the Pauli (Heisenberg) representation and adds no new physics. It gives three practical classifications in the units used by the rest of the qang project: which gates conserve Hamming weight, and therefore admit the qg symmetry witness and filter; which algorithms are read out from a few qg values and which need the full outcome histogram; and where reading qg values is strictly worse than the standard readout. Three readout identities are, as far as we know, not stated in this form elsewhere: the per-qubit polar bias after Grover iterations, Simon's secret as the unique non-trivial Z-parity equal to one, and quantum counting as a single qg_X . We then restate every gate and algorithm with the local radius of each qubit, $r_q^2 = qg_X^2 + qg_Y^2 + qg_Z^2$, the radius of the sphere on which the qubit lives (area $4\pi r_q^2$). For a pure global state $1 - r_q^2$ is the known one-tangle; for a noisy state it mixes entanglement and noise. In a state of fixed Hamming weight $r_q = |qg_Z|$, so the radius is read from Z-basis shots alone, and under equal T_1 the qg filter restores it exactly: in a pre-registered simulation the filtered deficit equals the noiseless tangle to 3×10^{-15} , while the unfiltered readout reports a false deficit of 0.60–0.95 on product states. Every identity is checked numerically and pinned by tests in the open-source library **qang** (version 0.6.4, `pip install qang`), module `qang.formulation`.

1 Why write gates and algorithms in qg units

The qang project describes a qubit by measurement-level quantities, the polar bias $qg_Z = \langle \sigma_z \rangle$ and its companions [1], and builds diagnostics on them: a register-mean witness and a shot-level filter for circuits that conserve a Hamming weight [2], and decoders that read relaxation from the final readout [3]. Those tools apply to some circuits and not to others. The question of this note is the bookkeeping one: for the gates and algorithms that people actually use, where do the qg tools apply, and what does a qg readout give? We answer it with the Pauli representation, which is standard [4, 5], stated in qg units and checked numerically line by line.

2 Framework

State. For one qubit, $\rho = \frac{1}{2}(I + qg_X X + qg_Y Y + qg_Z Z)$; for n qubits,

$$\rho = \frac{1}{2^n} \sum_{P \in \{I, X, Y, Z\}^{\otimes n}} qg_P P, \quad qg_P = \text{Tr}(\rho P), \quad (1)$$

and the purity is $\text{Tr}\rho^2 = 2^{-n} \sum_P qg_P^2$. Strings with one non-identity letter give the local qg values; the others are correlations, such as qg_{ZZ} .

Measurement. A Z-basis measurement gives, from the same shots, every diagonal value $qg_{Z_S} = \sum_y p(y)(-1)^{S \cdot y}$ for every subset S of qubits; for one qubit $P(0) = (1 + qg_Z)/2$. Values with X or Y need a basis rotation first.

Gates. A gate U maps qg values as

$$qg'_P = qg_{U^\dagger P U}. \quad (2)$$

For a Clifford gate, $U^\dagger P U$ is a signed Pauli string and the gate permutes qg values with signs; for a non-Clifford gate it is a combination, and the gate mixes them. Convention: the leftmost letter of a string is qubit 0, which is the control of CX and CZ (and the controls of Toffoli and Fredkin).

Weight conservation. If U maps each Hamming-weight sector into itself, the register mean $\overline{qg_Z} = 1 - 2N/n$ stays fixed, and the qg witness and filter apply.

Local radius. Each qubit's local qg values (qg_X, qg_Y, qg_Z) are a point in the unit ball. Its squared distance from the centre,

$$r_q^2 = qg_X^2 + qg_Y^2 + qg_Z^2 = 2 \text{Tr}\rho_q^2 - 1, \quad (3)$$

is 1 for a pure qubit and 0 for a maximally mixed one; the qubit lives on a sphere of area $4\pi r_q^2$, and the surface deficit is $4\pi(1 - r_q^2)$. The radius carries the same information as the area and is linear in the purity, so we use r_q^2 . If the whole register is pure, $1 - r_q^2 = 4 \det \rho_q$ is the one-tangle of qubit q with the rest (for two qubits the squared concurrence [6, 7]), and its mean over qubits is the Meyer–Wallach global entanglement Q [8, 9]. If the register is mixed, the deficit mixes entanglement with noise and cannot tell them apart. Reading r_q^2 for every qubit needs three settings for the whole register (all X, all Y, all Z). From N shots the plug-in $\hat{q}^2$ is biased upwards by $(1 - q^2)/N$; $(N\hat{q}^2 - 1)/(N - 1)$ is exactly unbiased.

3 Fifteen gates

Table 1: Action of each gate on qg values, Eq. (2); $c = \cos \theta$, $s = \sin \theta$. Values not listed are unchanged. Last column: effect on the local radii, Eq. (3), for product inputs (maximal deficit $1 - r^2$ and an input that reaches it).

gate	action on qg values	reading	local radius
1. Pauli X	$(qg_X, qg_Y, qg_Z) \rightarrow (qg_X, -qg_Y, -qg_Z)$	flips the polar bias	kept
2. Pauli Y	$(qg_X, qg_Y, qg_Z) \rightarrow (-qg_X, qg_Y, -qg_Z)$	flips qg_Z and qg_X	kept
3. Pauli Z	$(qg_X, qg_Y, qg_Z) \rightarrow (-qg_X, -qg_Y, qg_Z)$	invisible in the Z basis	kept
4. Hadamard	$(qg_X, qg_Y, qg_Z) \rightarrow (qg_Z, -qg_Y, qg_X)$	swaps polar bias and qg_X	kept

gate	action on qg values	reading	local radius
5. Phase S	$(qg_X, qg_Y) \rightarrow (-qg_Y, qg_X)$	quarter turn on the equator	kept
$P(\varphi)$	$qg'_X = \cos \varphi qg_X - \sin \varphi qg_Y,$ $qg'_Y = \sin \varphi qg_X + \cos \varphi qg_Y$	same as $R_z(\varphi)$	kept
6. T ($\pi/8$)	$qg'_X = (qg_X - qg_Y)/\sqrt{2},$ $qg'_Y = (qg_X + qg_Y)/\sqrt{2}$	first gate that mixes: irrational qg values (“magic”)	kept
7. $R_x(\theta)$	$qg'_Z = c qg_Z + s qg_Y,$ $qg'_Y = c qg_Y - s qg_Z$	rotation in the YZ plane	kept
8. $R_y(\theta)$	$qg'_Z = c qg_Z - s qg_X,$ $qg'_X = c qg_X + s qg_Z$	from $ 0\rangle$: $qg_Z = \cos \theta$, the qang identity; the RQang gate is $R_y(\arccos qg_Z)$	kept
9. $R_z(\theta)$	$qg'_X = c qg_X - s qg_Y,$ $qg'_Y = s qg_X + c qg_Y$	invisible in the Z basis	kept
10. CNOT	$qg'_{XI} = qg_{XX}, qg'_{IZ} = qg_{ZZ},$ $qg'_{IY} = qg_{ZY}, qg'_{YI} = qg_{YX}$	turns a correlation into a local value: the target’s qg_Z becomes the parity qg_{ZZ} (syndrome extraction)	to 0 ($ +0\rangle$)
11. CZ	$qg'_{XI} = qg_{XZ}, qg'_{IX} = qg_{ZX},$ $qg'_{YI} = qg_{YZ}, qg'_{IY} = qg_{ZY}$	invisible in the Z basis; conserves weight	to 0 ($ ++\rangle$)
12. SWAP	$qg'_{ab} = qg_{ba}$	relabels; conserves weight	exchanged
13. iSWAP	$qg'_{ZI} = qg_{IZ}, qg'_{XI} = -qg_{ZY},$ $qg'_{YI} = qg_{ZX}, qg'_{IX} = -qg_{YZ},$ $qg'_{IY} = qg_{XZ}$	swap with phase; conserves weight	to 0 ($ ++\rangle$)
14. Toffoli	$qg'_{IIZ} = \frac{1}{2}(qg_{IIZ} + qg_{IZZ} + qg_{ZIZ} - qg_{ZZZ})$	mixes inside the diagonal sector	to 0 ($ 1+0\rangle$)
15. Fredkin	$qg'_{IZI} = \frac{1}{2}(qg_{IZI} + qg_{ZZI} + qg_{IIZ} - qg_{ZIZ})$	controlled swap; conserves weight	to 0 ($ +01\rangle$)

Three observations organize the table.

- **Clifford or not.** X, Y, Z, H, S, CNOT, CZ, SWAP and iSWAP permute qg values with signs. T, generic rotations, Toffoli and Fredkin mix them: one string opens into two (T) or four (Toffoli, Fredkin) per application. That growth is what makes non-Clifford circuits hard to simulate [4].
- **Z-basis blindness.** Z, S, T, R_z , $P(\varphi)$ and CZ leave every qg_Z fixed; their effect appears only after a gate that moves the equator to the poles. This is, gate by gate, the Z-basis blind spot of the qg metrics [1].
- **Weight conservation.** Among multi-qubit gates, CZ, SWAP, iSWAP and Fredkin conserve Hamming weight; CNOT and Toffoli do not. A circuit built only from weight-conserving gates and R_z rotations has zero sector exposure, and in our simulations the qg filter then removes almost all relaxation error; basis changes around two-qubit gates let about half of it through [2].
- **Radius.** A one-qubit gate rotates the sphere and keeps every radius, also for a mixed qubit: it moves the point, never its distance from the centre. SWAP exchanges radii.

CNOT, CZ, iSWAP, Toffoli and Fredkin can take a product input to the centre ($r = 0$, deficit 1) and can bring it back. On basis states none of them creates a deficit: there they act as classical permutations, and the deficit needs a superposition at the input.

4 Fourteen algorithms

For each: the readout identity, what reading it in qg gives, and what it does not.

1. Deutsch–Jozsa [10]. With f as a phase oracle, $P(0^n) = 2^{-n} \sum_S qg_{ZS}$: the all-zero probability is the mean of all Z-parities. In qg : f is constant if and only if all n local qg_Z equal +1; one value below 1 already means balanced. No correlation is needed. *Radius.* A constant f leaves every qubit on the surface ($r = 1$); a non-linear balanced f on four qubits leaves $r^2 = 0.25$ on each, so any deficit also certifies “balanced” (a linear balanced f still gives a product state, $r = 1$).

2. Bernstein–Vazirani [11]. The output is the product state $|s\rangle$, so $qg_Z^{(i)} = (-1)^{s_i}$: each secret bit is the sign of its qubit’s polar bias. Under noise each bit is decided separately, with its own interval. *Radius.* $r = 1$ on every qubit: the output is a product, and the local values are the whole answer.

3. Simon [12]. The output is uniform over $\{y : y \cdot s = 0\}$, so $qg_{ZS} = 1$ for $S \in \{0, s\}$ and 0 otherwise: the secret is the unique non-trivial Z-parity equal to one. This describes the answer but does not find it efficiently, since there are 2^n parities; the efficient route remains a linear system from $O(n)$ samples. *Radius.* For $n = 3$, $s = 110$, every output qubit sits at the centre ($r = 0$): it is maximally entangled with the oracle register, and nothing about s is local.

4. Shor [13]. The quantum part estimates the phase of an order r . If r is a power of two, the local values show it: for $r = 4$ with six counting qubits, $qg_Z = (0, 0, 1, 1, 1, 1)$. For $r = 6$ the local qg_Z lie between 0 and 0.33 and do not determine r ; the information is in the correlations, and the full bit string plus continued fractions is needed. This is a negative result for local readout. *Radius.* For $r = 4$ the counting qubits have $r^2 = (0, 0, 1, 1, 1, 1)$: the pure qubits carry no information about the order, the two at the centre carry it. For $r = 6$ every counting qubit has $r^2 \leq 0.11$: the register is almost maximally mixed qubit by qubit.

5. Grover [14]. With one marked item x^* among $N = 2^n$ and success probability P after k iterations,

$$qg_Z^{(i)} = (-1)^{x_i^*} \frac{NP - 1}{N - 1}. \quad (4)$$

The marked item is the sign pattern of the local polar biases, and their common magnitude measures success. Reading it this way costs more shots than the histogram, about $1/P^2$ against $1/P$, as we found in a noisy simulation; the identity is a diagnostic, not a better readout. *Radius.* With $n = 5$ the radius of every qubit goes $1 \rightarrow 0.82 \rightarrow 0.66 \rightarrow 0.84 \rightarrow 0.998$ over $k = 0, \dots, 4$: Grover entangles the register on the way and disentangles it at the optimum, where the answer becomes local again.

6. Phase estimation [15] and 11. phase kickback. A control in $|+\rangle$ acting on an eigenstate of U with eigenvalue $e^{i\varphi}$ ends with

$$qg_X + i qg_Y = e^{i\varphi}, \quad qg_Z \text{ unchanged.} \quad (5)$$

Kickback writes the phase into the control's equatorial angle. Iterative phase estimation reads qg_X and qg_Y for $2^k\varphi$, one bit at a time; it is the natural qg form of phase estimation, and the same mechanism drives Deutsch–Jozsa and Bernstein–Vazirani. *Radius*. A Hadamard test of U on $|\psi\rangle$ leaves the control with $r^2 = |\langle\psi|U|\psi\rangle|^2$: exactly 1 for an eigenstate, and below 1 otherwise (0.81 for $\varphi = 0.9$ on $|+\rangle$). The radius tests the eigenstate assumption on which phase estimation rests, and its deficit is the control's entanglement with the target.

7. HHL [16]. The output $|x\rangle \propto A^{-1}|b\rangle$ cannot be read out in full; what HHL delivers is expectation values $\langle x|M|x\rangle$, that is qg values of the solution register, and its success probability is the ancilla's polar bias, $P(1) = (1 - qg_Z^{\text{anc}})/2$. The qg form states plainly what HHL produces: qg values, not the vector [17]. We check this readout for a 2×2 matrix by exact linear algebra, not with a circuit. *Radius*. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1, 0)$ the ancilla has $r^2 = 0.32$: it is entangled with the solution register whenever b spans eigenvectors with different eigenvalues.

8. Quantum Fourier transform. The QFT of a basis state $|x\rangle$ is a product of equatorial qubits: qubit l has $qg_Z = 0$ and $qg_X + i qg_Y = e^{2\pi i x/2^l}$. The QFT moves the information from polar biases (the bits of x) to equatorial angles, which the Z basis does not see; hence it is always followed by a basis change before measurement. *Radius*. $r = 1$ on every qubit: the QFT of a basis state only rotates each point on the surface, from the poles to the equator.

9. VQE [18]. The energy is linear in qg values, $E = \sum_P c_P qg_P$, grouped by measurement basis. This is the main use of the qg tools so far: witness, filter, symmetry checks and pole-damped optimization [1, 2]. *Radius*. The local part of the energy is bounded by the radii, $|\sum_\alpha c_{q,\alpha} qg_\alpha^{(q)}| \leq \|c_q\| r_q$. For the two-qubit singlet ($XX + YY + ZZ$) both radii are 0 and all the energy is in correlations; for the two-site Ising chain at $h = 1$, $r^2 = 0.8$.

10. QAOA [19]. For MaxCut, $C = \sum_{(i,j)} w_{ij}(1 - qg_{Z_i Z_j})/2$: the whole cost comes from ZZ correlations in the same shots. With a weight-conserving XY mixer the witness and filter apply [2]. *Radius*. On a ring of four at the best $p = 1$ angles every qubit has $r^2 = 0.25$, which is why the cost lives in ZZ. With an XY mixer the state has fixed weight and $r_q = |qg_Z^{(q)}|$.

12. Quantum counting [20]. A Hadamard test of the Grover operator G on the uniform state $|s\rangle$ leaves the control with

$$qg_X = \langle s|G|s\rangle = 1 - \frac{2M}{N} \implies M = \frac{N(1 - qg_X)}{2}. \quad (6)$$

Counting solutions is measuring one qg_X : with $N = 16$ and three marked items, $qg_X = 0.625$ and $M = 3$. The precision follows single-qubit statistics; full counting by phase estimation gains precision with more counting qubits. *Radius*. Here $qg_Y = qg_Z = 0$, so $r = |qg_X| = |1 - 2M/N|$ ($r^2 = 0.39$ in the example) and the deficit $4(M/N)(1 - M/N)$ is the control's entanglement with the register, largest at $M = N/2$.

13. Quantum walks [21]. A continuous-time single walker with hopping $(XX + YY)/2$ conserves the number of excitations, so the register mean stays at $\overline{qg_Z} = 1 - 2/n$ at all times (checked for $n = 5$): single-walker walks live in the weight-one sector, where the witness and filter apply as in Heisenberg dynamics. Discrete coined walks do not conserve weight. *Radius*. In a fixed-weight state $qg_X = qg_Y = 0$ on every qubit, so $r_q = |qg_Z^{(q)}|$ and the deficit is $4p_q(1 - p_q)$, with p_q the occupation of site q : the walker's spread is its entanglement (r^2 from 0.02 to 0.87 at $t = 2$ for $n = 5$).

14. Machine learning: Q-SVM [22], Q-PCA [23]. The fidelity kernel is an all-zero probability, $k(x, x') = P(0^n) = 2^{-n} \sum_S qg_{Z_S}$, and for product angle encodings it reduces to $\prod_i (1 + \vec{q}_i \cdot \vec{q}_i')/2$, the qang transition probability. In Q-PCA the purity and spectrum of ρ depend on squared qg values, $\text{Tr}\rho^2 = 2^{-n} \sum_P qg_P^2$. *Radius.* Product angle encodings keep $r = 1$; under noise the kernel diagonal becomes $k(x, x) = \prod_q (1 + r_q^2)/2$, a direct readout of the radii.

5 Readout classes

Table 2: Which qg values suffice to read out each algorithm, and the local radius of the readout qubits (`qang.formulation.algorithm_radii`). The pattern: the closer the readout qubits are to the surface, the more local the readout; when they sit near the centre the answer is in the correlations.

sufficient readout	algorithms	radius of the readout qubits
local qg_Z one ancilla's qg_X, qg_Y	Deutsch–Jozsa, Bernstein–Vazirani phase kickback, iterative phase estimation, quantum counting (Hadamard test), HHL success	1 (product output) $ \langle U \rangle $; 1 only for an eigenstate
few-qubit correlations	QAOA (ZZ), VQE (Hamiltonian strings)	below 1; 0 for a singlet
all parities or the histogram	Simon, Shor with general order, Grover (local readout exists, at $\sim 1/P^2$ shots), fidelity kernels	near 0 (Simon, Shor); Grover returns to 1

6 The radius after the qg filter

The radius is an entanglement measure only when the register is pure, and noise is what makes it mixed. Circuits that conserve Hamming weight give a way out. In a state of fixed weight every local qg_X and qg_Y vanishes, so $r_q = |qg_Z^{(q)}|$ and the deficit $1 - qg_Z^2$ is read from Z-basis shots alone. Under equal T_1 the qg filter keeps the shots that stayed in the sector, and the kept state is the noiseless one: on that sector the no-decay branch of T_1 is a multiple of the identity (RESEARCH_NOTES §77). The filtered deficit should therefore be the noiseless one-tangle, and the unfiltered one should not.

We tested this with predictions committed before the run (RESEARCH_NOTES §98, `examples/qg_radius_witness_qg.py`): random weight-conserving circuits of reconfigurable beam splitters on 4 and 6 qubits, weight 1 to $n/2$, depth 12 to 48, $\gamma = 0.02$ per qubit per layer, 1000 and 10000 shots, the estimator $1 - (N\hat{q}^2 - 1)/(N - 1)$ with and without the filter.

- Under equal T_1 the exact filtered deficit equals the noiseless tangle to 3×10^{-15} ; the unfiltered one errs by 0.06–0.37 on average.
- On product states (basis inputs, idle circuits) the filtered deficit is exactly 0 under equal T_1 , unequal T_1 and dephasing; the unfiltered readout reports a deficit of 0.60–0.95 on every excited qubit: false entanglement from decay alone.
- From shots, the filter has the lower error in 30 of 30 configurations under equal T_1 (7.6 to 1005 times lower mean squared error), 30 of 30 under unequal T_1 and 24 of 30 with dephasing, where the filtered deficit is biased (up to 0.31).
- One prediction failed: we expected the unfiltered error to be smaller at half filling, by a first-order cancellation near occupation 1/2. It is larger (0.177 against 0.074 for $n = 4$, depth 12).

The radius therefore joins the qg toolbox with a clear domain: in weight-conserving circuits under equal T_1 the filter turns the surface deficit into an entanglement measure, from the same Z-basis shots as the readout.

7 What is new and what is not

The Pauli representation of states, the action of gates by conjugation, and the stabilizer formalism are known [4, 5], and so are the algorithms [10–16, 18–23]. The contribution is organizational and practical: one language, the qg units of the qang project, for three questions. (i) Which gates conserve weight and admit the qg witness and filter. (ii) Which algorithms are read out from a few qg values and which need the histogram. (iii) Where qg readout is worse (Grover, Shor with general order, Simon). The readout forms in Eqs. (4) and (6), and Simon’s secret as the unique unit parity, are elementary but, to our knowledge, not stated in this form elsewhere. The local radius, its relation to purity, the one-tangle and the Meyer–Wallach measure are known [6–9]; what we add is its table over the fifteen gates and fourteen algorithms in qg units, the observation that in a fixed-weight state it is read from Z-basis shots, and the test showing that the qg filter makes it an entanglement measure under equal T_1 . None of this is a quantum advantage.

8 Limitations

(i) Ideal, noiseless states throughout; noise enters only through the cited qg studies. (ii) The identities are checked on small instances ($n \leq 6$). (iii) The HHL readout is checked by exact linear algebra for a 2×2 matrix, not with a circuit. (iv) Shot costs are quoted from a separate simulation (Grover) or are the standard single-qubit statistics. (v) The radius deficit is an entanglement measure only for a pure register; after the filter this holds exactly under equal T_1 alone, and the §98 test is a simulation.

9 Code availability

The library `qang` (MIT license) is on PyPI (`pip install qang`, version 0.6.4) and at <https://github.com/Viny2030/qang>. The module `qang.formulation` gives the gate matrices, the rule of Eq. (2) (`conjugate`, `gate_table`, `apply_gate`), qg values of a state (`qg_values`), one readout function per algorithm, and the radius functions (`radius_profile`, `radius_deficit`, `sphere_area`, `meyer_wallach`, `gate_radius_class`, `algorithm_radii`; unbiased estimates in `qang.statistics.radius2_estimate`); `tests/test_formulation.py` pins every identity, and `notebooks/qang_inicio_formulacion.ipynb` runs them in Colab. For example:

```
from qang import formulation as F
bell = F.apply_gate(F.apply_gate(F.qg_values([1, 0, 0, 0]),
                                     F.kron(F.H, F.I2)), F.CX)
# qg_XX = 1, qg_YY = -1, qg_ZZ = 1, all local values 0
F.radius_profile(bell) # [0.0, 0.0]: both qubits at the centre
```

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